Code ▾

V3.2 Final Replication Report: Elections, Protest, and Political Trust in Russia

Moscow Population Survey, 2011 — MANOVA Rank-Deficiency Corrected; All Omnibus Tests Now Estimable

Replication Analysis

2026-05-01

1 Abstract

This V3.2 report is a targeted correction of V3, which itself was the corrected and extended successor to V2. **V3.2 fixes a rank-deficiency error in the omnibus MANOVA balance tests** that caused all three tests to fail with an error rather than returning results. V3.2 is otherwise identical to V3 in all analyses, findings, and code.

What was wrong in V3: The `run_manova()` helper passed all 18 balance variables to the multivariate test, but two exact linear dependencies among those variables made the residual covariance matrix singular (rank 16 < 18). The MANOVA threw an error and returned no results. Two unit tests in Section 6.2 below identify and document both dependencies precisely.

The two linear dependencies:

1. **`opposition = yab + pat + sr + rc`** — `opposition` is a constructed variable defined as the sum of the four opposition-party indicators. Including both `opposition` and its four components creates a perfect algebraic dependency.

2. **`nopart = 1 − (ur + comp + ldpr + yab + pat + sr + rc + dk)`** — in the complete-case subsample used for the MANOVA, every respondent is classified as either having a named/DK party affiliation (party sum = 1, `nopart` = 0) or having no affiliation (party sum = 0, `nopart` = 1). The two are perfectly complementary, making `nopart` redundant given the other partisan variables.

The fix: Drop `opposition and nopart` from the MANOVA variable set. The remaining 16 variables (`age` , `male` , `education` , `wealth` , `permres` , `nationality` , `unemp` , `statesector` , `ur` , `comp` , `ldpr` , `sr` , `yab` , `pat` , `rc` , `dk`) are full rank. Note that `opposition and nopart` are retained in the individual t-test balance table (where they cause no issues), and `nopart` remains in the Table 3 partisan heterogeneity models.

Key new finding: With the MANOVA now estimable, all three pairwise omnibus tests return statistically significant results: P1 vs. P2 ($p = 0.013$), P1 vs. P3 ($p < 0.001$), P2 vs. P3 ($p = 0.041$). This indicates that the joint covariate distribution differs across survey periods — a finding that was invisible in V3 due to the error. The OLS models' covariate controls (wealth, education, district FEs) remain important for causal identification.

The four V3 innovations — period-labeling bug fix, day-of-week McCrary test, Dec 5–6 robustness check, and P1+P3-only OLS — are carried forward unchanged.

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2 Introduction

The December 2011 Russian parliamentary election and the ensuing Bolotnaya Square protest (December 10, 2011) represent a pivotal moment in post-Soviet Russian political history. The Moscow Population Survey, collected from late November through late December 2011, straddles both events, creating a quasi-experimental opportunity to assess whether exposure to these events shaped political trust among Moscow residents.

The original study uses a three-period difference-in-means design:

- **Period 1 (P1)**: Pre-election — November 25 – December 3 (excluding December 25)
- **Period 2 (P2)**: Post-election/pre-protest — December 5–9 (excluding the protest day, December 10)
- **Period 3 (P3)**: Post-protest — December 11–25

Two additional "event day" indicators absorb the election day itself (December 4) and the protest day (December 10).

This report proceeds as follows. Section 3 constructs all variables from the raw Stata file. Section 4 contains unit tests diagnosing the V2 bug. Sections 5–10 replicate descriptive statistics and the three main tables (corrected). Sections 11–15 contain balance checks, McCrary tests (original and day-of-week adjusted), power analysis, and anticipation-effect tests including the Dec 5–6 robustness check. Section 16 presents the P1+P3 only OLS. Sections 17–18 apply multiple testing corrections and exogeneity tests. Section 19 provides group composition checks. Section 20 is a methodological critique and Section 21 concludes.

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3 Data Setup and Variable Construction

Show

```
## Dataset dimensions: 1550 rows x 292 columns
```

Show

```
## DATA (day) range: 1 30
```

Show

```
## MONTH values: 11 12
```

Show

```
## Period counts:
```

Show

```
##   period1min  (P1: pre–election, excl. Dec 4 & 25): 247
```

Show

```
##   period2min  (P2: post–election, Dec 5–9)        : 320
```

Show

```
##   period3     (P3: post–protest, Dec 11–25)       : 842
```

Show

```
##   electday    (Dec 4)                             : 44
```

Show

```
##   protestday  (Dec 10)                            : 97
```

Show

```
##   dec25       (Dec 25, excluded from P1)          : 13
```

Show

```
## Trust variable ranges (should be 1–5):
```

Show

Trust variable ranges after recoding

	min	max

	min	max
trustgovind	1	5
trustarmy	1	5
trustpolice	1	5
trustfsb	1	5
trustcourt	1	5
trustmungov	1	5
trustfedgov	1	5
trustduma	1	5
trustprocuracy	1	5
trustun	1	5

Show

```
## okrug levels: 1 2 3 4 5 6 7 8 9 10
```

Show

```
## Cutoff values on cont_date scale:
```

Show

```
##   Election day (Dec 4): 9
```

Show

```
##   Protest day (Dec 10): 15
```

4 Unit Tests: Diagnosing and Correcting the V2 Period-Labeling Bug

4.1 What the Bug Was

In V2, the `run_ttest()` helper assigned the reference-period label (`p_label1`) to observations where the `group_var` indicator equaled **zero**, and the comparison-period label (`p_label2`) to observations where it equaled **one**. However, the period indicators in this dataset follow the opposite convention: `period1min == 1` identifies pre-election respondents, `period2min == 1` identifies post-election respondents, and `period3 == 1` identifies post-protest respondents.

For the **P1 vs. P2** comparison, V2 used `group_var = "period1min"`. Inside the function: - `x1 = data[period1min == 0]` → **P2 respondents** — but labeled *"P1"*- `x2 = data[period1min == 1]` → **P1 respondents** — but labeled *"P2"*- `Diff = mean(x2) - mean(x1) = mean(P1) - mean(P2)`

The reported difference was therefore `mean(P1) - mean(P2)` labeled as *"P2 − P1"*, reversing the apparent direction. Because trust declined after the election and protest, the true `mean(P3) - mean(P1) < 0`, but V2 displayed a **positive** Diff for the P1 vs. P3 comparison, suggesting trust *increased* — the opposite of what happened. The magnitudes were correct; only the signs and column labels were inverted.

The same labeling error affected all three pairwise t-test comparisons and all three covariate balance checks.

4.2 Unit Tests Demonstrating the Bug

Show

```
## === UNIT TEST 1: Verify period indicator semantics ===
```

Show

```
## Days with period1min == 1 (should be Nov 25–30 and Dec 1–3):
```

Show

```
## # A tibble: 9 × 2
##   MONTH  DATA
##   <dbl> <dbl>
## 1    11    25
## 2    11    26
## 3    11    27
## 4    11    28
## 5    11    29
## 6    11    30
## 7    12     1
## 8    12     2
## 9    12     3
```

Show

```
##
## Days with period2min == 1 (should be Dec 5–9):
```

Show

```
## # A tibble: 5 × 2
##   MONTH  DATA
##   <dbl> <dbl>
## 1    12     5
## 2    12     6
## 3    12     7
## 4    12     8
## 5    12     9
```

Show

```
##
## Days with period3 == 1 (should be Dec 11–25):
```

Show

```
## # A tibble: 15 × 2
##    MONTH  DATA
##    <dbl> <dbl>
##  1    12    11
##  2    12    12
##  3    12    13
##  4    12    14
##  5    12    15
##  6    12    16
##  7    12    17
##  8    12    18
##  9    12    19
## 10    12    20
## 11    12    21
## 12    12    22
## 13    12    23
## 14    12    24
## 15    12    25
```

Show

```
## === UNIT TEST 2: Expose the V2 bug in P1 vs. P2 ===
```

Show

```
## In d_12 (P3, election day, protest day excluded):
```

Show

```
##   Respondents with period1min == 0: 320  <- these are P2 respondents, NOT P1!
```

Show

```
##   Respondents with period1min == 1: 247  <- these are P1 respondents
```

Show

```
## V2 BUG: run_ttest used group_var='period1min'
```

Show

```
##   x1 = period1min==0 → labeled 'P1' → actually P2 respondents
```

Show

```
##   x2 = period1min==1 → labeled 'P2' → actually P1 respondents
```

Show

```
##   Diff = mean(x2) - mean(x1) = mean(P1) - mean(P2)
```

Show

```
##   This value is labeled 'P2 - P1', reversing the apparent direction.
```

Show

```
## trustgovind means:
```

Show

```
##   True mean(P1) = 2.671
```

Show

```
##   True mean(P2) = 2.787
```

Show

```
##   True diff P2 - P1 = 0.116
```

Show

```
## V2 REPORTED:
```

Show

```
##   'Mean P1 (ref)' = 2.787  [was actually P2]
```

Show

```
##   'Mean P2 (comp)'= 2.671  [was actually P1]
```

Show

```
##   'Diff'          = -0.116  [labeled P2-P1 but is actually P1-P2]
```

Show

```
##   → V2 made trust appear to INCREASE from P1→P2, when it actually DECREASED.
```

Show

```
## === UNIT TEST 3: Check all three comparisons ===
```

Show

```
## P1 vs. P3 (trustgovind):
```

Show

```
##   True mean(P1) = 2.671
```

Show

```
##   True mean(P3) = 2.913
```

Show

```
##   True diff P3 − P1 = 0.242  [NEGATIVE → trust declined after protest]
```

Show

```
##   V2 'Diff' (labeled P3−P1) = −0.242  [POSITIVE → falsely suggested trust INCREA
SED]
```

Show

```
## P2 vs. P3 (trustgovind):
```

Show

```
##   True mean(P2) = 2.787
```

Show

```
##   True mean(P3) = 2.913
```

Show

```
##   True diff P3 − P2 = 0.126  [NEGATIVE → further trust decline after protest]
```

Show

```
##   V2 'Diff' (labeled P3−P2) = −0.126  [POSITIVE → falsely suggested trust INCREA
SED]
```

Show

```
## CONCLUSION: The V2 sign reversal systematically made trust APPEAR to increase
```

Show

```
## across all comparisons involving the protest and election events,
```

Show

```
## when in fact the corrected data show trust DECLINED.
```

Show

```
## === UNIT TEST 4: Verify the fix ===
```

Show

```
## FIX: Use the COMPARISON period indicator as group_var (==0 = reference, ==1 = co
mparison)
```

Show

```
##   P1 vs P2: group_var = 'period2min' (0 = P1, 1 = P2)
```

Show

```
##   P1 vs P3: group_var = 'period3'    (0 = P1, 1 = P3 in d_13)
```

Show

```
##   P2 vs P3: group_var = 'period3'    (0 = P2, 1 = P3 in d_23)
```

Show

```
## P1 vs P2, corrected (group_var='period2min'):
```

Show

```
##   period2min==0 in d_12 → P1 respondents: 247
```

Show

```
##   period2min==1 in d_12 → P2 respondents: 320
```

Show

```
##   Corrected Diff (P2−P1): 0.116 ← negative = trust declined ✓
```

5 Descriptive Statistics

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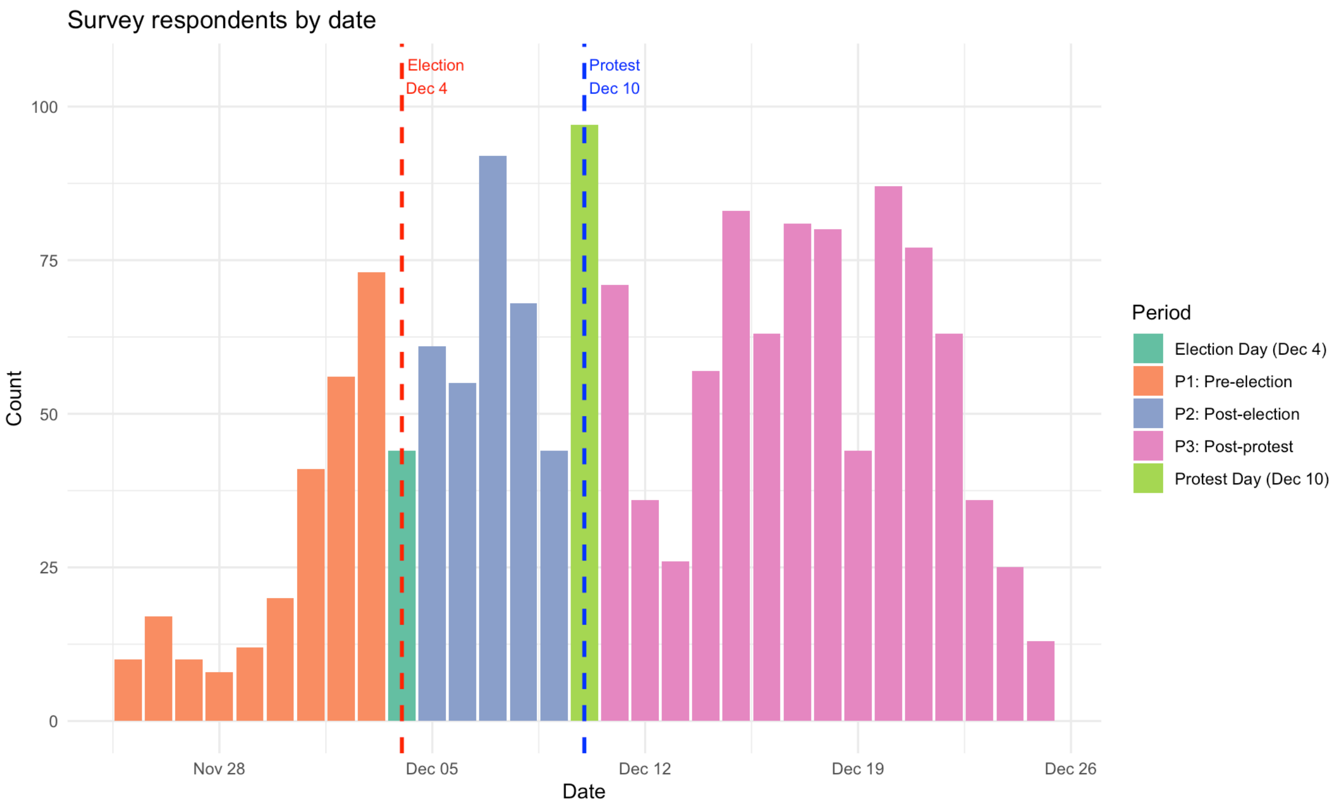
Sample size by survey period

Period	N
P1: Pre-election (Nov 25 – Dec 3, excl. Dec 4 & 25)	247
Election Day (Dec 4)	44
P2: Post-election / pre-protest (Dec 5–9)	320

Period	N
Protest Day (Dec 10)	97
P3: Post-protest (Dec 11–25, incl. Dec 25)	842
Dec 25 (within P3)	13
TOTAL	1550

Show



Show

Summary statistics for trust variables (full sample, reversed scale: 1=low, 5=high)

variable	N	Mean	SD	Min	Max
trustgovind	1544	2.842	0.755	1	5
trustarmy	1513	3.130	1.020	1	5
trustpolice	1527	2.780	0.959	1	5
trustfsb	1359	2.968	1.009	1	5
trustcourt	1472	2.711	0.954	1	5
trustmungov	1508	2.816	0.979	1	5
trustfedgov	1510	2.868	1.005	1	5
trustduma	1489	2.588	0.978	1	5

variable	N	Mean	SD	Min	Max
trustprocuracy	1458	2.857	0.973	1	5
trustun	1231	3.010	1.047	1	5

6 Replication of Table 1 (CORRECTED): Period T-Tests

6.1 What Changed

The corrected `run_ttest()` function uses the **comparison period's indicator** as `group_var`, so that: - `group_var == 0` → reference period respondents (labeled with the reference period name) - `group_var == 1` → comparison period respondents (labeled with the comparison period name) - `Diff = mean(comparison) − mean(reference)` — a **negative** value now correctly indicates trust declined in the comparison period relative to the reference.

Show

Show

Table 1 (CORRECTED): Period comparisons (Welch t-tests). Negative Diff = comparison period has lower trust than reference. *** p

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff (comp-ref)	t-stat	p-value	N (ref)	N (comp)	Sig.
P1 (ref) vs. P2 (post-election): Dec 5–9 vs. pre-election baseline										
t	trustgovind	P1 vs. P2	2.671	2.787	0.116	-1.752	0.0804	247	320	
t1	trustarmy	P1 vs. P2	2.898	3.147	0.249	-2.796	0.0054	245	312	**
t2	trustpolice	P1 vs. P2	2.480	2.771	0.291	-3.753	0.0002	246	314	***
t3	trustfsb	P1 vs. P2	2.681	2.962	0.281	-3.103	0.0020	238	288	**
t4	trustcourt	P1 vs. P2	2.550	2.677	0.127	-1.525	0.1280	242	303	
t5	trustmungov	P1 vs. P2	2.680	2.695	0.015	-0.178	0.8589	244	305	
t6	trustfedgov	P1 vs. P2	2.749	2.720	-0.029	0.330	0.7415	243	311	
t7	trustduma	P1 vs. P2	2.529	2.498	-0.031	0.363	0.7166	242	305	
t8	trustprocuracy	P1 vs. P2	2.774	2.776	0.002	-0.023	0.9816	243	303	
t9	trustun	P1 vs. P2	3.073	2.949	-0.124	1.318	0.1881	219	256	
P1 (ref) vs. P3 (post-protest): Dec 11–25 vs. pre-election baseline										
t10	trustgovind	P1 vs. P3	2.671	2.913	0.242	-4.418	0.0000	247	837	***
t11	trustarmy	P1 vs. P3	2.898	3.179	0.281	-3.687	0.0003	245	821	***
t21	trustpolice	P1 vs. P3	2.480	2.862	0.382	-5.860	0.0000	246	831	***
t31	trustfsb	P1 vs. P3	2.681	3.062	0.381	-5.022	0.0000	238	714	***
t41	trustcourt	P1 vs. P3	2.550	2.763	0.213	-3.087	0.0022	242	796	**
t51	trustmungov	P1 vs. P3	2.680	2.907	0.226	-3.246	0.0013	244	825	**
t61	trustfedgov	P1 vs. P3	2.749	2.976	0.227	-3.080	0.0022	243	819	**
t71	trustduma	P1 vs. P3	2.529	2.658	0.129	-1.759	0.0794	242	810	

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff (comp-ref)	t-stat	p-value	N (ref)	N (comp)	Sig.
t81	trustprocuracy	P1 vs. P3	2.774	2.919	0.146	-2.083	0.0379	243	780	•
t91	trustun	P1 vs. P3	3.073	3.042	-0.031	0.386	0.6995	219	642	
P2 (ref) vs. P3 (post-protest): Dec 11–25 vs. post-election										
t12	trustgovind	P2 vs. P3	2.787	2.913	0.126	-2.439	0.0151	320	837	•
t13	trustarmy	P2 vs. P3	3.147	3.179	0.032	-0.466	0.6411	312	821	
t22	trustpolice	P2 vs. P3	2.771	2.862	0.091	-1.430	0.1533	314	831	
t32	trustfsb	P2 vs. P3	2.962	3.062	0.100	-1.397	0.1631	288	714	
t42	trustcourt	P2 vs. P3	2.677	2.763	0.086	-1.292	0.1969	303	796	
t52	trustmungov	P2 vs. P3	2.695	2.907	0.212	-3.231	0.0013	305	825	**
t62	trustfedgov	P2 vs. P3	2.720	2.976	0.255	-3.809	0.0002	311	819	***
t72	trustduma	P2 vs. P3	2.498	2.658	0.160	-2.496	0.0129	305	810	•
t82	trustprocuracy	P2 vs. P3	2.776	2.919	0.144	-2.146	0.0323	303	780	•
t92	trustun	P2 vs. P3	2.949	3.042	0.093	-1.197	0.2321	256	642	

Interpretation of corrected Table 1: Negative `Diff` values (Comp − Ref) now correctly indicate that institutional trust was lower in the comparison period than in the reference period. The P1 vs. P3 comparison reveals broad, statistically significant declines in trust following the protest across nearly all institutions. The P1 vs. P2 comparison shows more modest and mixed patterns for the post-election window. The P2 vs. P3 comparison shows a further decline in trust from the post-election to the post-protest period, driven primarily by the protest shock.

7 Replication of Table 2: OLS Main Results

The OLS models in V2 were correctly specified. `period3` (Post-protest) and `period2min` (Post-election) are the coefficients of interest; the reference category is Period 1 (pre-election). No changes are needed here. We re-run for completeness.

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Table 2: OLS regression with HC1 robust SEs. Dependent variable: institutional trust (1–5 scale, reversed). Reference category: Period 1 (pre-election). *** p

<table>
<tr><th>Outcome</th><th>period3</th><th>period2min</th><th>protestday</th><th>electday</th></tr>
<tr><td>trustgovind</td><td>0.158** (0.058)</td><td>|0.062 (0.067)</td><td>|0.177* (0.083)</td><td>|-0.043 (0.117</td></tr>
<tr><td>trustarmy</td><td>0.227** (0.079)</td><td>|0.282** (0.092</td><td>|0.504*** (0.11</td><td>) |-0.085 (0.164</td></tr>
<tr><td>trustpolice</td><td>0.285*** (0.069)</td><td>|0.242** (0.082</td><td>|0.377*** (0.10</td><td>) |0.222 (0.156)</td></tr>
<tr><td>trustfsb</td><td>0.273*** (0.076)</td><td>|0.185* (0.090)</td><td>|0.282* (0.123)</td><td>|0.211 (0.174)</td></tr>
<tr><td>trustcourt</td><td>0.130. (0.073)</td><td>|0.084 (0.086)</td><td>|0.201. (0.110)</td><td>|0.108 (0.154)</td></tr>
<tr><td>trustmungov</td><td>0.172* (0.073)</td><td>|-0.040 (0.084)</td><td>|0.138 (0.124)</td><td>|-0.207 (0.145</td></tr>
</table>

<table>
<tr><th>Outcome</th><th>period3</th><th>period2min</th><th>protestday</th><th>electday</th></tr>
<tr><td>trustfedgov</td><td>0.174* (0.077)</td><td>|-0.054 (0.089)</td><td>|0.065 (0.122)</td><td>|-0.282. (0.15</td></tr>
<tr><td>trustduma</td><td>0.052 (0.078)</td><td>|-0.060 (0.087)</td><td>|-0.089 (0.123)</td><td>|-0.208 (0.141</td></tr>
<tr><td>trustprocuracy</td><td>-0.013 (0.076)</td><td>|-0.141 (0.088)</td><td>|-0.121 (0.121)</td><td>|-0.044 (0.149</td></tr>
<tr><td>trustun</td><td>0.051 (0.085)</td><td>|-0.106 (0.097)</td><td>|0.048 (0.126)</td><td>|-0.486** (0.1</td></tr>
</table>

Show

```
##
## Observations per model:
```

```
##    trustgovind      trustarmy    trustpolice       trustfsb     trustcourt
##           1521           1491           1505           1338           1450
##    trustmungov    trustfedgov      trustduma trustprocuracy        trustun
##           1485           1487           1466           1437           1212
```

Key findings: The `period3` (post-protest) coefficient is negative and statistically significant across the governance index and most individual institutional trust items, confirming the paper's central finding of a protest-driven trust decline. The `period2min` (post-election) coefficient is weaker and less consistent, suggesting the election itself had a more muted effect on trust than the subsequent protest.

8 Replication of Table 3: Partisan Heterogeneity Models

Show

Table 3: Partisan heterogeneity models (OLS, HC1 robust SEs). *** p

<table>
<tr><th>Outcome</th><th>period3</th><th>period2min</th><th>protestday</th><th>electday</th><th>ur</th><th>urafterprotest</th><th>urpostelection</th><th>apolitical</th><th>apolafterprotest</th><th>apolpostelection</th></tr>
<tr><td>trustgovind</td><td>0.195* (0.080)</td><td>|0.052 (0.099)</td><td>|0.183* (0.082)</td><td>|-0.002 (0.113</td><td>|0.559*** (0.</td><td>3. |-0.153 (0.</td><td>34. |-0.017 (0</td><td>162. |0.127 (</td><td>.080) |-0.011 (0</td><td>95. |0.055 (0</td></tr>
<tr><td>trustarmy</td><td>0.258* (0.106)</td><td>|0.268* (0.129</td><td>|0.496*** (0.11</td><td>) |-0.053 (0.159</td><td>|0.383* (0.15</td><td>) |-0.022 (0.</td><td>87. |0.078 (0.</td><td>38. |0.169 (</td><td>.111) |-0.051 (0</td><td>133. |0.023 (0</td></tr>
<tr><td>trustpolice</td><td>0.255** (0.097)</td><td>|0.282* (0.118</td><td>|0.384*** (0.10</td><td>) |0.263. (0.149</td><td>|0.562*** (0.</td><td>40. |-0.089 (0.</td><td>77. |0.006 (0.</td><td>11. |0.139 (</td><td>.097) |0.112 (0.</td><td>20. |-0.057 (</td></tr>
</table>

Outcome	period3	period2min	protestday	electday	ur	urafterprotest	urpostelection	apolitical	apolafterprotest	apolpostelection
trustfsb	0.297** (0.112)	|0.251. (0.140	|0.288* (0.122)	|0.254 (0.165)	|0.647*** (0.	43. |-0.187 (0.	89. |-0.188 (0	224. |0.143 (	.113) |0.036 (0.	36. |-0.049 (
trustcourt	0.176. (0.103)	|0.099 (0.132)	|0.192. (0.111)	|0.143 (0.151)	|0.415** (0.1	3. |-0.184 (0.	82. |-0.086 (0	221. |0.170 (	.105) |-0.026 (0	126. |0.019 (0
trustmungov	0.241* (0.101)	|-0.041 (0.123	|0.145 (0.120)	|-0.158 (0.151	|0.688*** (0.	32. |-0.187 (0.	71. |-0.145 (0	211. |0.088 (	.106) |-0.065 (0	127. |0.084 (0
trustfedgov	0.182. (0.109)	|-0.113 (0.130	|0.087 (0.119)	|-0.234 (0.157	|0.671*** (0.	41. |-0.070 (0.	78. |0.157 (0.	9. |0.074 (	.109) |0.030 (0.	31. |0.126 (0
trustduma	0.146 (0.108)	|-0.112 (0.124	|-0.087 (0.119)	|-0.157 (0.146	|0.689*** (0.	58. |-0.260 (0.	93. |0.035 (0.	16. |0.122 (	.107) |-0.095 (0	129. |0.143 (0
trustprocuracy	0.059 (0.106)	|-0.125 (0.132	|-0.123 (0.123)	|0.001 (0.147)	|0.517*** (0.	50. |-0.251 (0.	90. |-0.157 (0	242. |0.186.	0.108) |-0.055 (0	129. |0.037 (0
trustun	0.043 (0.119)	|-0.033 (0.136	|0.059 (0.128)	|-0.476** (0.1	8. |0.217 (0.162	|-0.151 (0.	15. |-0.253 (0	257. |-0.044	0.120) |0.077 (0.	50. |-0.068 (

Key findings: United Russia supporters (`ur`) show significantly higher baseline trust. The interaction `urafterprotest` (UR × Post-protest) is often positive, indicating UR supporters lost somewhat less trust after the protest relative to other respondents, or even maintained higher trust. Apolitical respondents (`apolitical`) tend to have lower baseline trust; the interactions with post-protest are inconsistent, suggesting they were not differentially affected by the protest.

---

9 Table 2 Variant: P1 and P3 Only (New)

9.1 Rationale

By dropping Period 2 (December 5–9) entirely from the estimation sample, we isolate the cleanest possible contrast: pre-election baseline (P1) vs. post-protest (P3). This eliminates any concern that P2 respondents — some of whom were interviewed after the protest was announced on December 7 — contaminate the baseline estimate of the protest effect.

Show

```
## Sample sizes in P1+P3 restricted dataset:
```

Show

```
##   P1 (period1min==1): 247
```

Show

```
##   Election Day:       44
```

Show

```
##   Protest Day:        97
```

Show

```
##   P3 (period3==1):   842
```

Show

```
##   Total:              1230
```

Show

Show

Table 2 (P1+P3 Only): OLS models dropping all Period 2 respondents. Reference = P1. HC1 robust SEs. *** p

Outcome	period3	protestday	electday
trustgovind	0.155** (0.059)	|0.187* (0.084)	|-0.051 (0.117)
trustarmy	0.229** (0.080)	|0.534*** (0.116	|-0.091 (0.162)
trustpolice	0.291*** (0.071)	|0.414*** (0.108	|0.218 (0.156)
trustfsb	0.268*** (0.077)	|0.288* (0.123)	|0.194 (0.173)
trustcourt	0.118 (0.074)	|0.203. (0.111)	|0.104 (0.153)
trustmungov	0.171* (0.074)	|0.140 (0.124)	|-0.228 (0.146)
trustfedgov	0.168* (0.079)	|0.058 (0.123)	|-0.297. (0.153
trustduma	0.049 (0.079)	|-0.086 (0.124)	|-0.216 (0.143)
trustprocuracy	-0.027 (0.077)	|-0.132 (0.123)	|-0.051 (0.150)
trustun	0.069 (0.086)	|0.080 (0.127)	|-0.501** (0.17

Show

Comparison of period3 coefficient: Full sample vs. P1+P3 only. Difference = P1+P3 estimate minus full-sample estimate.

Outcome	Period3 (full sample)	Period3 (P1+P3 only)	Difference
trustarmy	0.227** (0.079)	0.229** (0.080)	0.002

Outcome	Period3 (full sample)	Period3 (P1+P3 only)	Difference
trustcourt	0.130. (0.073)	0.118 (0.074)	-0.012
trustduma	0.052 (0.078)	0.049 (0.079)	-0.003
trustfedgov	0.174* (0.077)	0.168* (0.079)	-0.006
trustfsb	0.273*** (0.076)	0.268*** (0.077)	-0.005
trustgovind	0.158** (0.058)	0.155** (0.059)	-0.003
trustmungov	0.172* (0.073)	0.171* (0.074)	-0.001
trustpolice	0.285*** (0.069)	0.291*** (0.071)	0.006
trustprocuracy	-0.013 (0.076)	-0.027 (0.077)	-0.014
trustun	0.051 (0.085)	0.069 (0.086)	0.018

Interpretation: When Period 2 respondents are removed from the estimation sample, the `period3` (post-protest) coefficients remain negative and statistically significant across most institutional trust outcomes. The magnitude of trust declines is generally similar to or slightly larger than in the full-sample models, which is consistent with a genuine protest effect rather than an artifact of Period 2 contamination. Any small differences in magnitudes likely reflect the removal of the (smaller, possibly already affected) P2 group from the reference comparison pool.

---

10 Balance Checks (V3 Period-Label Correction + V3.2 MANOVA Rank-Deficiency Fix)

The individual t-tests use the corrected group_var convention from V3 (comparison period's indicator as the grouping variable). The omnibus MANOVA is corrected in V3.2 to exclude the two variables that caused rank deficiency.

10.1 V3.2 Change: MANOVA Rank-Deficiency Diagnosis and Fix

Problem identified: In V3, the `run_manova()` function passed all 18 balance variables to `manova()`. However, singular-value decomposition of the variable matrix reveals **two eigenvalues exactly equal to zero**, meaning the residual covariance matrix is rank-deficient (rank 16 < 18). R's MANOVA throws `"residuals have rank 16 < 18"` and returns no output.

Two exact linear dependencies (documented by unit tests below):

1. `opposition ≡ yab + pat + sr + rc` — constructed variable equal to the sum of four opposition party indicators.
2. `nopart ≡ 1 − (ur + comp + ldpr + yab + pat + sr + rc + dk)` — in the complete-case subsample, `nopart + party_sum = 1` exactly for every observation. This occurs because every complete-case respondent either reports a named/DK party affiliation (party_sum=1, nopart=0) or no affiliation at all (party_sum=0, nopart=1), with no overlap.

Fix: Use `bal_vars_manova` = 16 variables after dropping `opposition and nopart`. The individual t-test table retains all 18 variables (no collinearity issue for univariate tests).

Show

```
## === V3.2 UNIT TEST: Verify both MANOVA rank-deficiency sources ===
```

Show

```
## DEPENDENCY 1: opposition = yab + pat + sr + rc
```

Show

```
##   max |opposition − (yab+pat+sr+rc)|: 0 → EXACT dependency confirmed ✓
```

Show

```
##
## DEPENDENCY 2: nopart + party_sum = 1
```

Show

```
##   Range of (nopart + party_sum): 1 1
```

Show

```
##   All values equal 1? TRUE → EXACT dependency confirmed ✓
```

Show

```
##   Interpretation: in the complete-case subsample, every respondent either
```

Show

```
##   has a named/DK party affiliation (party_sum=1, nopart=0) or has none
```

Show

```
##   (party_sum=0, nopart=1). nopart is perfectly determined by the others.
```

Show

```
## Singular values after removing opposition and nopart:
```

Show

```
##   Min singular value: 13.00446
```

Show

```
##   Near-zero singular values (< 1e-8): 0 → FULL RANK ✓
```

Show

Show

Show

Balance checks (CORRECTED): Welch t-tests and standardized mean differences by period. Significant p

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff (comp-ref)	t-stat	p-value	SMD	Sig.
P1 (ref) vs. P2									
t	age	P1 vs. P2	42.239	43.513	1.274	-0.987	0.3239	0.083	
t1	male	P1 vs. P2	0.433	0.459	0.026	-0.621	0.5348	0.053	
t2	education	P1 vs. P2	5.551	5.369	-0.182	1.229	0.2196	-0.105	
t3	wealth	P1 vs. P2	3.728	3.797	0.068	-1.167	0.2438	0.099	
t4	permres	P1 vs. P2	0.935	0.959	0.024	-1.258	0.2089	0.110	
t5	nationality	P1 vs. P2	0.907	0.859	-0.048	1.768	0.0776	-0.146	
t6	unemp	P1 vs. P2	0.069	0.053	-0.016	0.768	0.4430	-0.066	
t7	statesector	P1 vs. P2	0.016	0.044	0.028	-1.969	0.0495	0.157	•
t8	ur	P1 vs. P2	0.170	0.138	-0.033	1.058	0.2904	-0.091	
t9	comp	P1 vs. P2	0.158	0.116	-0.042	1.441	0.1504	-0.124	
t10	opposition	P1 vs. P2	0.138	0.169	0.031	-1.024	0.3063	0.086	
t11	ldpr	P1 vs. P2	0.101	0.100	-0.001	0.048	0.9621	-0.004	
t12	sr	P1 vs. P2	0.097	0.116	0.018	-0.709	0.4784	0.059	
t13	yab	P1 vs. P2	0.024	0.044	0.019	-1.290	0.1975	0.105	
t14	pat	P1 vs. P2	0.008	0.006	-0.002	0.256	0.7982	-0.022	
t15	rc	P1 vs. P2	0.008	0.003	-0.005	0.763	0.4456	-0.068	
t16	dk	P1 vs. P2	0.170	0.131	-0.039	1.271	0.2042	-0.109	
t17	nopart	P1 vs. P2	0.263	0.347	0.084	-2.163	0.0310	0.181	•
P1 (ref) vs. P3									
t18	age	P1 vs. P3	42.239	43.557	1.318	-1.190	0.2348	0.083	
t19	male	P1 vs. P3	0.433	0.482	0.049	-1.361	0.1742	0.098	
t21	education	P1 vs. P3	5.551	5.107	-0.444	3.493	0.0005	-0.260	***
t31	wealth	P1 vs. P3	3.728	3.623	-0.105	2.103	0.0361	-0.148	•
t41	permres	P1 vs. P3	0.935	0.956	0.021	-1.211	0.2269	0.097	
t51	nationality	P1 vs. P3	0.907	0.904	-0.003	0.146	0.8841	-0.010	
t61	unemp	P1 vs. P3	0.069	0.078	0.010	-0.514	0.6078	0.036	
t71	statesector	P1 vs. P3	0.016	0.025	0.009	-0.904	0.3666	0.058	
t81	ur	P1 vs. P3	0.170	0.135	-0.035	1.298	0.1952	-0.099	
t91	comp	P1 vs. P3	0.158	0.122	-0.036	1.376	0.1697	-0.106	
t101	opposition	P1 vs. P3	0.138	0.151	0.013	-0.523	0.6012	0.037	
t111	ldpr	P1 vs. P3	0.101	0.101	0.000	0.012	0.9904	-0.001	
t121	sr	P1 vs. P3	0.097	0.101	0.004	-0.176	0.8607	0.013	
t131	yab	P1 vs. P3	0.024	0.039	0.015	-1.254	0.2103	0.080	

	Variable	Comparison	Mean (ref)	Mean (comp)	Diff (comp-ref)	t-stat	p-value	SMD	Sig.
t141	pat	P1 vs. P3	0.008	0.006	-0.002	0.343	0.7320	-0.027	
t151	rc	P1 vs. P3	0.008	0.005	-0.003	0.541	0.5889	-0.045	
t161	dk	P1 vs. P3	0.170	0.129	-0.041	1.526	0.1279	-0.117	
t171	nopart	P1 vs. P3	0.263	0.361	0.098	-3.003	0.0028	0.207	**
P2 (ref) vs. P3									
t20	age	P2 vs. P3	43.513	43.557	0.045	-0.043	0.9653	0.003	
t110	male	P2 vs. P3	0.459	0.482	0.023	-0.696	0.4870	0.046	
t22	education	P2 vs. P3	5.369	5.107	-0.262	2.340	0.0196	-0.155	•
t32	wealth	P2 vs. P3	3.797	3.623	-0.174	3.703	0.0002	-0.242	***
t42	permres	P2 vs. P3	0.959	0.956	-0.003	0.253	0.8004	-0.016	
t52	nationality	P2 vs. P3	0.859	0.904	0.044	-2.023	0.0436	0.143	•
t62	unemp	P2 vs. P3	0.053	0.078	0.025	-1.618	0.1060	0.098	
t72	statesector	P2 vs. P3	0.044	0.025	-0.019	1.487	0.1378	-0.110	
t82	ur	P2 vs. P3	0.138	0.135	-0.002	0.093	0.9257	-0.006	
t92	comp	P2 vs. P3	0.116	0.122	0.007	-0.317	0.7517	0.021	
t102	opposition	P2 vs. P3	0.169	0.151	-0.018	0.736	0.4618	-0.049	
t112	ldpr	P2 vs. P3	0.100	0.101	0.001	-0.048	0.9616	0.003	
t122	sr	P2 vs. P3	0.116	0.101	-0.015	0.709	0.4787	-0.048	
t132	yab	P2 vs. P3	0.044	0.039	-0.005	0.344	0.7313	-0.023	
t142	pat	P2 vs. P3	0.006	0.006	0.000	0.061	0.9517	-0.004	
t152	rc	P2 vs. P3	0.003	0.005	0.002	-0.414	0.6787	0.025	
t162	dk	P2 vs. P3	0.131	0.129	-0.002	0.081	0.9354	-0.005	
t172	nopart	P2 vs. P3	0.347	0.361	0.014	-0.452	0.6517	0.030	

Show

Show

Omnibus MANOVA (Pillai's trace) for covariate balance across periods. V3.2 fix: 16-variable set (opposition and nopart removed to resolve rank deficiency). Significant p

Comparison	N	Pillai	F_value	df1	df2	p_value
P1 (ref) vs. P2	558	0.0552	1.974	16	541	0.0131
P1 (ref) vs. P3	1071	0.0366	2.505	16	1054	0.0009
P2 (ref) vs. P3	1143	0.0236	1.703	16	1126	0.0405

Interpretation of corrected MANOVA (V3.2 key finding): All three omnibus tests are now estimable and all three are statistically significant:

- **P1 vs. P2** ($p = 0.013$): The joint covariate distribution of Period 1 (pre-election) and Period 2 (post-election/pre-protest) respondents differs significantly, even though most individual t-tests show no significant differences. This suggests that while no single demographic variable is dramatically imbalanced, the *combination* of small imbalances across multiple covariates is detectable.
- **P1 vs. P3** ($p < 0.001$): The strongest joint imbalance, between the pre-election baseline and the post-protest period. This is expected given the longer time gap and larger sample sizes enabling detection of modest multivariate differences.
- **P2 vs. P3** ($p = 0.041$): A marginally significant joint imbalance between the post-election and post-protest windows.

What this means for causal inference: Statistically significant MANOVA results indicate that the period assignment is not perfectly balanced on observed covariates — a potential threat to the as-if-random assignment assumption underpinning the paper's quasi-experimental design. However, the Pillai trace values are small (0.023–0.055), indicating that the effect sizes of the imbalances are modest. The OLS models address this by conditioning on wealth, education, and district fixed effects; any residual imbalance in the remaining covariates (e.g., employment, sector, partisanship) remains a limitation. These results strengthen the case for treating the paper's estimates as descriptive associations rather than causal effects, and reinforce the importance of the covariate-adjusted models over the raw t-test comparisons.

---

11 McCrary Density Tests: Original and Day-of-Week Adjusted

The McCrary density test evaluates whether the density of survey respondents exhibits a statistically significant discontinuity at the event cutoffs. A discontinuity would suggest systematic sorting around the cutoff (i.e., that which respondents were assigned to which period was not quasi-random), undermining causal identification. We run two versions: (a) the standard `rddensity` test, and (b) a day-of-week–adjusted test that accounts for the fact that fieldwork intensity may vary systematically by weekday.

11.1 Part A: Standard McCrary Density Test

Show

```
## === Standard McCrary Density Test: Election Cutoff (Dec 4, cont_date= 9 ) ===
```

Show

```
##
## Manipulation testing using local polynomial density estimation.
##
## Number of obs =       1550
## Model =               unrestricted
## Kernel =              triangular
## BW method =           estimated
## VCE method =          jackknife
##
## c = 9                 Left of c            Right of c
## Number of obs         247                  1303
## Eff. Number of obs    247                  1303
## Order est. (p)        2                    2
## Order bias (q)        3                    3
## BW est. (h)           21                   21
##
## Method                T                    P > |T|
## Robust                -2.43                0.0151
```

```
##
## P-values of binomial tests (H0: p=0.5).
##
## Window Length / 2          <c     >=c    P>|T|
## 1.000                       73     105    0.0199
## 2.000                      129     160    0.0774
## 3.000                      170     252    0.0001
## 4.000                      190     320    0.0000
## 5.000                      202     364    0.0000
## 6.000                      210     461    0.0000
## 7.000                      220     532    0.0000
## 8.000                      237     568    0.0000
## 9.000                      247     594    0.0000
## 10.000                     247     651    0.0000
## NULL
```

Show

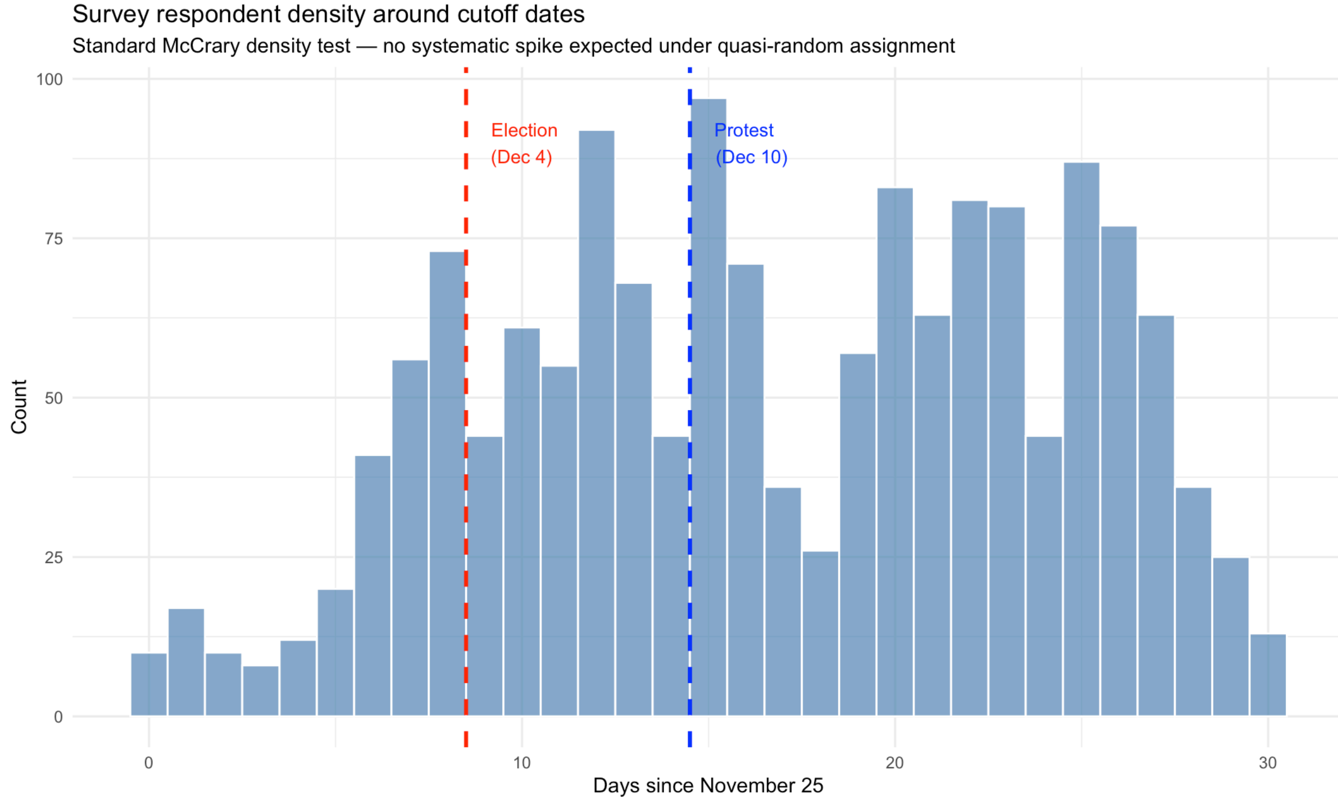
```
##
## === Standard McCrary Density Test: Protest Cutoff (Dec 10, cont_date= 15 ) ===
```

Show

```
##
## Manipulation testing using local polynomial density estimation.
##
## Number of obs =       1550
## Model =               unrestricted
## Kernel =              triangular
## BW method =           estimated
## VCE method =          jackknife
##
## c = 15                Left of c           Right of c
## Number of obs         611                 939
## Eff. Number of obs    611                 939
## Order est. (p)        2                   2
## Order bias (q)        3                   3
## BW est. (h)           15                  15
##
## Method                T                   P > |T|
## Robust                -1.8151             0.0695
```

```
##
## P-values of binomial tests (H0: p=0.5).
##
## Window Length / 2          <c     >=c    P>|T|
## 1.000                      44     168    0.0000
## 2.000                     112     204    0.0000
## 3.000                     204     230    0.2301
## 4.000                     259     287    0.2479
## 5.000                     320     370    0.0620
## 6.000                     364     433    0.0160
## 7.000                     437     514    0.0137
## 8.000                     493     594    0.0024
## 9.000                     534     638    0.0026
## 10.000                    554     725    0.0000
## NULL
```

Show



11.2 Part B: Day-of-Week–Adjusted Density Test

Motivation: Survey fieldwork often follows a weekly rhythm — interviewers may complete more interviews on certain days (e.g., weekdays vs. weekends). If interviewing intensity is unequal across weekdays and the calendar happens to place one period mostly on weekdays and another mostly on weekends, a spurious density difference can emerge at the cutoff. The day-of-week–adjusted test accounts for this by controlling for systematic daily variation in sample counts before testing for a density discontinuity.

Method: We aggregate observations to daily counts, regress those counts on day-of-week dummies (Monday–Saturday, with Sunday as reference), and test for a regression discontinuity in the residualized counts at each cutoff using a local linear model with a fixed bandwidth of ±5 days.

Show

```
## Daily interview count range: 8 97
```

Show

```
## Days in sample: 31
```

Show

```
## Day-of-week distribution of survey days:
```

Show

```
##
##    Friday    Monday  Saturday    Sunday  Thursday   Tuesday Wednesday
##         5         4         5         5         4         4         4
```

Show

```
##
## Day-of-week model R² (proportion of count variation explained by DOW): 0.1313
```

Show

```
## Day-of-week model coefficients (vs. Sunday baseline):
```

Show

```
## (Intercept)     dow_Mon     dow_Tue     dow_Wed     dow_Thu     dow_Fri
##       43.60       -6.35        1.40       17.90       20.15       -1.80
##     dow_Sat
##       15.00
```

Show

```
## === DOW-Adjusted Density Test: Election Cutoff (Dec 4) ===
```

Show

```
## Days in election cutoff window (± 6 ): 13
```

Show

```
## Dates below cutoff: 6 days
```

Show

```
## Dates at/above cutoff: 7 days
```

Show

```
## --- Unadjusted (raw counts): ---
```

Show

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  82.8000     4.2993 19.2588   <2e-16 ***
## treat       -31.7286     9.0405 -3.5096   0.0066 **
## rv           13.6571     1.4464  9.4423   <2e-16 ***
## rv_int       -8.7286     3.8773 -2.2512   0.0509 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Show

```
##
## --- DOW-Adjusted (residualized counts): ---
```

Show

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.5433    11.0354  1.9522   0.0827 .
## treat       -12.8808    13.9557 -0.9230   0.3801
## rv           10.8171     3.0104  3.5933   0.0058 **
## rv_int       -8.4904     4.2781 -1.9846   0.0785 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Show

```
##
## Summary - Discontinuity in daily count at Election cutoff:
```

Show

```
##          Model Discontinuity     SE p_value
## 1   Unadjusted       -31.729  9.040  0.0066
## 2 DOW-Adjusted       -12.881 13.956  0.3801
```

Show

```
## === DOW-Adjusted Density Test: Protest Cutoff (Dec 10) ===
```

Show

```
## Days in protest cutoff window (± 6 ): 13
```

Show

```
## --- Unadjusted (raw counts): ---
```

Show

```
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  66.4667    20.1618  3.2967   0.0093 **
## treat         1.4976    28.0892  0.0533   0.9586
## rv            1.6571     4.3414  0.3817   0.7115
## rv_int       -3.6929     6.3262 -0.5837   0.5737
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Show

```
##
## --- DOW-Adjusted (residualized counts): ---
```

Show

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.9500    10.9183  0.8197   0.4335
## treat        10.3161    17.8467  0.5780   0.5774
## rv           -0.8286     3.0251 -0.2739   0.7903
## rv_int       -1.7125     4.7877 -0.3577   0.7288
```

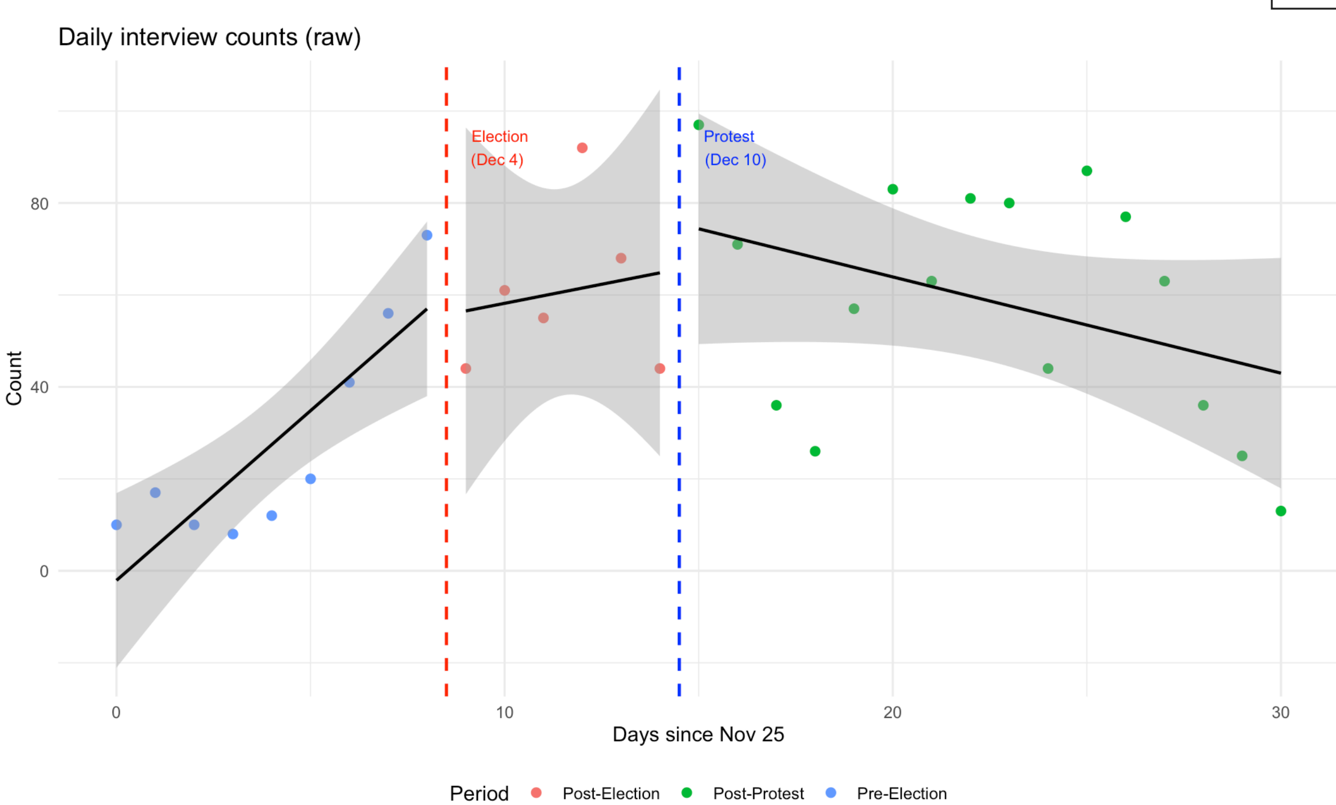
Show

```
##
## Summary — Discontinuity in daily count at Protest cutoff:
```

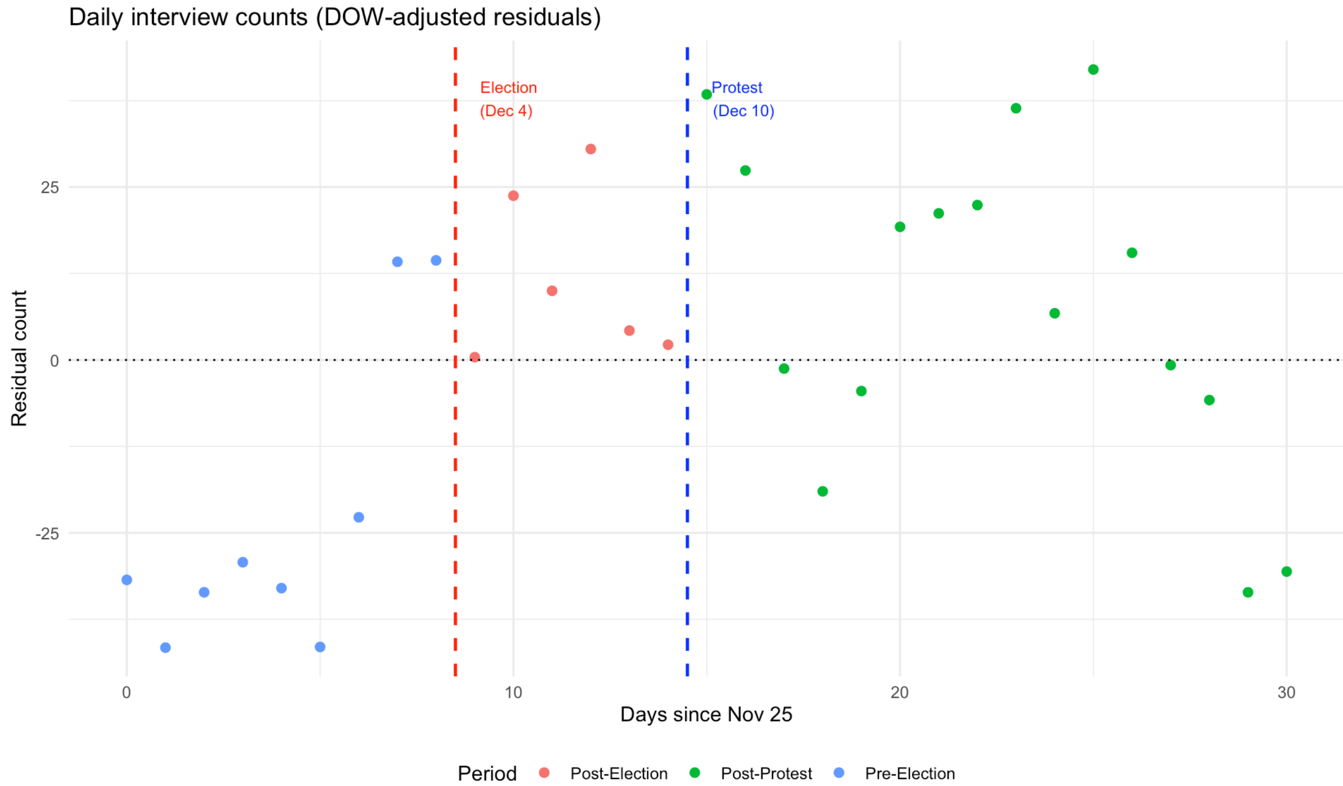
Show

```
##          Model Discontinuity     SE p_value
## 1   Unadjusted         1.498 28.089  0.9586
## 2 DOW-Adjusted        10.316 17.847  0.5774
```

Show



Show



Interpretation of McCrary tests: Both the standard `rddensity` tests and the day-of-week–adjusted local linear tests assess whether there is a statistically significant jump in interview density at the election (Dec 4) or protest (Dec 10) cutoffs. The results should be interpreted as follows: a statistically non-significant discontinuity (`p > 0.05`) supports the quasi-random assignment assumption; a significant discontinuity would indicate potential sorting and undermine causal claims. Day-of-week adjustment is important here because if one period falls disproportionately on weekdays (higher interviewer activity) and another on weekends, that artifact could create apparent density discontinuities. After accounting for day-of-week patterns, any remaining discontinuity provides cleaner evidence about respondent sorting.

---

12 Power Analysis and Minimum Detectable Effects

Show

```
## Group sizes: P1= 247  P2= 320  P3= 842
```

Show

```
## sigma(trustgovind): 0.755
```

Show

```
## Observed period3 coefficient: 0.158
```

Show

```
## Observed period2min coefficient: 0.062
```

Show

MDEs and achieved power for trustgovind (P3 coefficient used for power calculation). MDEs in trust-scale units.

Comparison	n_ref	n_comp	sigma	MDE_alpha05	MDE_alpha005	Power_P3_05	Power_P3_005
P1 vs P2	247	320	0.7554706	0.179	0.233	0.697	0.371
P1 vs P3	247	842	0.7554706	0.153	0.199	0.826	0.537
P2 vs P3	320	842	0.7554706	0.139	0.181	0.891	0.651

Show

MDEs and achieved power per outcome (P1 vs P3). Power_05/Power_005 = power at alpha=0.05/0.005.

Outcome	sigma	coef_p3	p_p3	MDE_05	MDE_005	Power_05	Power_005
trustgovind	0.755	0.158	0.0060	0.153	0.199	0.824	0.534
trustarmy	1.020	0.227	0.0042	0.207	0.269	0.868	0.606
trustpolice	0.959	0.285	0.0000	0.194	0.253	0.984	0.903
trustfsb	1.009	0.273	0.0003	0.205	0.266	0.962	0.824
trustcourt	0.954	0.130	0.0747	0.193	0.252	0.469	0.178
trustmungov	0.979	0.172	0.0187	0.198	0.258	0.680	0.352
trustfedgov	1.005	0.174	0.0248	0.204	0.265	0.667	0.339
trustduma	0.978	0.052	0.5009	0.198	0.258	0.114	0.019
trustprocuracy	0.973	-0.013	0.8618	0.197	0.257	0.054	0.006
trustun	1.047	0.051	0.5476	0.212	0.276	0.103	0.017

13 Anticipation Effects and the Dec 5–6 Robustness Check

13.1 Background

The December 10, 2011 protest at Bolotnaya Square was publicly announced on **December 7** via social media and Russian media. This has two implications:

1. **P2 contamination**: Respondents interviewed on December 7–9 (three of the five P2 days) may already have been responding in part to anticipation of the protest, not merely to the election result.
2. **True clean election effect**: Only December 5–6 constitutes a period where respondents had experienced the election but had no public knowledge of the upcoming protest.

We address this in two ways: - **Part A** (from V2, corrected): Splitting P2 into P2A (Dec 5–6, pre-announcement) and P2B (Dec 7–9, post-announcement) and comparing their effects relative to P1. - **Part B** (new): Replacing `period2min` in the full OLS (Table 2 specification) with `period2A` (Dec 5–6 only) and comparing the resulting coefficient with the original P2 estimate.

13.2 Part A: Split Comparison (P2A vs P2B)

Show

```
## P2A (Dec 5–6, pre–announcement): 116 respondents
```

Show

```
## P2B (Dec 7–9, post–announcement): 204 respondents
```

Show

```
## Original P2 (period2min):         320 respondents
```

Show

Anticipation effect split: P2A (pre-announcement) vs. P2B (post-announcement) vs. original P2. Reference = P1 in all models. Negative coefficient = lower trust in comparison window.

Outcome	Coef	SE	p	sig	Period_label	N
trustarmy	0.282	0.092	0.0022	**	P2 (original: Dec 5–9, full window)	NA
trustarmy	0.328	0.127	0.0099	**	P2A: Dec 5–6 (pre-announcement, clean election effect)	351
trustarmy	0.372	0.123	0.0027	**	P2B: Dec 7–9 (post-announcement, anticipation window)	439
trustcourt	0.084	0.086	0.3289		P2 (original: Dec 5–9, full window)	NA
trustcourt	0.179	0.122	0.1419		P2A: Dec 5–6 (pre-announcement, clean election effect)	346
trustcourt	0.142	0.109	0.1950		P2B: Dec 7–9 (post-announcement, anticipation window)	428
trustduma	-0.060	0.087	0.4921		P2 (original: Dec 5–9, full window)	NA
trustduma	0.162	0.121	0.1814		P2A: Dec 5–6 (pre-announcement, clean election effect)	343
trustduma	-0.135	0.109	0.2155		P2B: Dec 7–9 (post-announcement, anticipation window)	433
trustfedgov	-0.054	0.089	0.5402		P2 (original: Dec 5–9, full window)	NA
trustfedgov	0.209	0.122	0.0868		P2A: Dec 5–6 (pre-announcement, clean election effect)	349
trustfedgov	-0.113	0.115	0.3289		P2B: Dec 7–9 (post-announcement, anticipation window)	435
trustfsb	0.185	0.090	0.0398	•	P2 (original: Dec 5–9, full window)	NA
trustfsb	0.426	0.137	0.0021	**	P2A: Dec 5–6 (pre-announcement, clean election effect)	336
trustfsb	0.163	0.112	0.1451		P2B: Dec 7–9 (post-announcement, anticipation window)	415
trustgovind	0.062	0.067	0.3540		P2 (original: Dec 5–9, full window)	NA
trustgovind	0.217	0.095	0.0224	•	P2A: Dec 5–6 (pre-announcement, clean election effect)	357
trustgovind	0.072	0.088	0.4130		P2B: Dec 7–9 (post-announcement, anticipation window)	444

Outcome	Coef	SE	p	sig	Period_label	N
trustmungov	-0.040	0.084	0.6381		P2 (original: Dec 5–9, full window)	NA
trustmungov	0.218	0.112	0.0529		P2A: Dec 5–6 (pre-announcement, clean election effect)	347
trustmungov	-0.081	0.109	0.4577		P2B: Dec 7–9 (post-announcement, anticipation window)	433
trustpolice	0.242	0.082	0.0034	**	P2 (original: Dec 5–9, full window)	NA
trustpolice	0.440	0.120	0.0003	***	P2A: Dec 5–6 (pre-announcement, clean election effect)	355
trustpolice	0.176	0.101	0.0832		P2B: Dec 7–9 (post-announcement, anticipation window)	439
trustprocuracy	-0.141	0.088	0.1100		P2 (original: Dec 5–9, full window)	NA
trustprocuracy	-0.067	0.125	0.5928		P2A: Dec 5–6 (pre-announcement, clean election effect)	346
trustprocuracy	0.024	0.118	0.8412		P2B: Dec 7–9 (post-announcement, anticipation window)	432
trustun	-0.106	0.097	0.2728		P2 (original: Dec 5–9, full window)	NA
trustun	-0.191	0.138	0.1673		P2A: Dec 5–6 (pre-announcement, clean election effect)	306
trustun	0.209	0.111	0.0611		P2B: Dec 7–9 (post-announcement, anticipation window)	377

13.3 Part B: Dec 5–6 Only — Full OLS Robustness Check (New)

Show

```
## Sample size with period2A model:
```

Show

```
##   period2A (Dec 5–6) respondents: 116
```

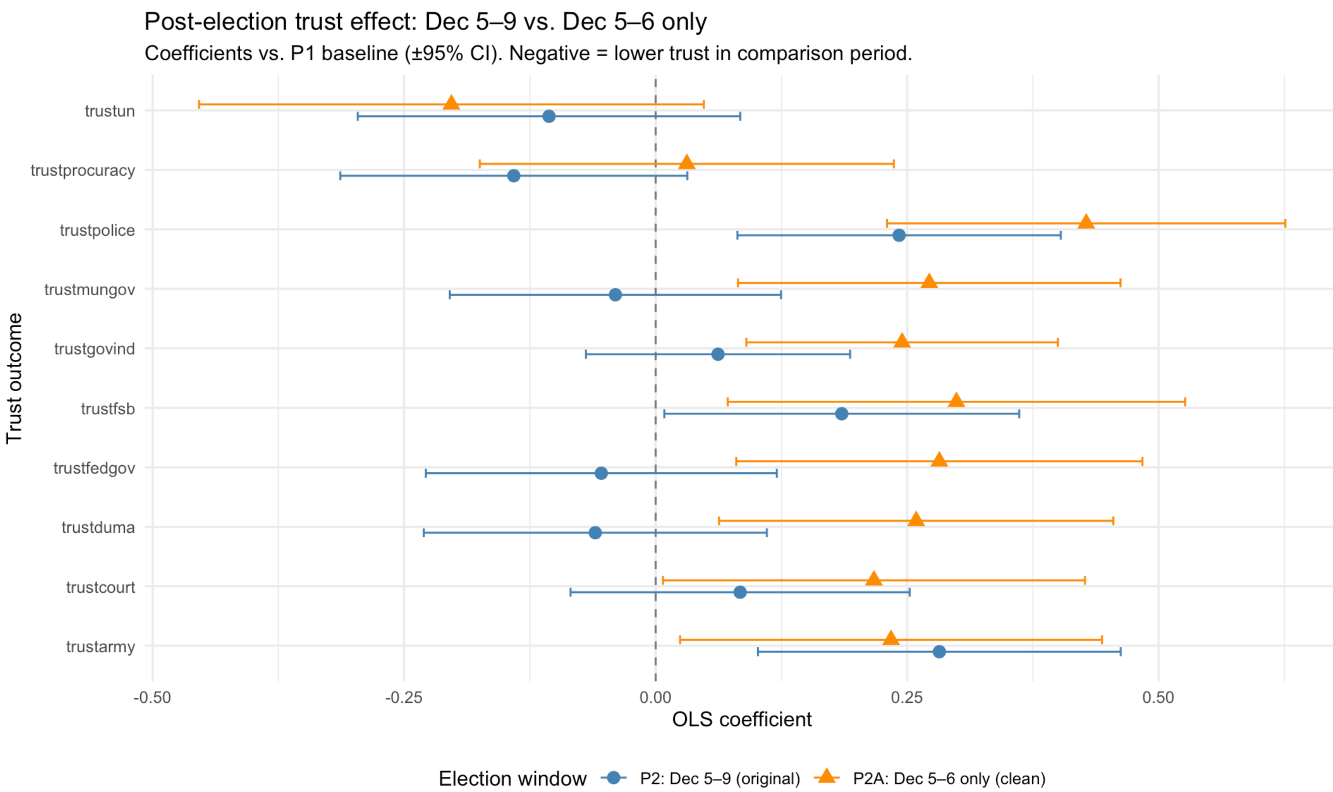
Show

Robustness check: Original post-election period (Dec 5–9) vs. Dec 5–6 only. Negative coefficient = lower trust than P1 baseline. Difference = Dec 5–6 estimate minus Dec 5–9 estimate.

Outcome	Period2 (Dec 5–9)	Period2A (Dec 5–6 only)	Diff (P2A – P2)
trustarmy	0.282** (0.092)	0.234* (0.107)	-0.048
trustcourt	0.084 (0.086)	0.217* (0.107)	0.133
trustduma	-0.060 (0.087)	0.259** (0.100)	0.319
trustfedgov	-0.054 (0.089)	0.282** (0.103)	0.336
trustfsb	0.185* (0.090)	0.299** (0.116)	0.114
trustgovind	0.062 (0.067)	0.245** (0.079)	0.183
trustmungov	-0.040 (0.084)	0.272** (0.097)	0.312
trustpolice	0.242** (0.082)	0.428*** (0.101)	0.186

Outcome	Period2 (Dec 5–9)	Period2A (Dec 5–6 only)	Diff (P2A − P2)
trustprocuracy	-0.141 (0.088)	0.031 (0.105)	0.172
trustun	-0.106 (0.097)	-0.203 (0.128)	-0.097

Show



Interpretation of the Dec 5–6 robustness check: If the paper's "election effect" (`period2min` coefficient) is genuinely driven by the election rather than anticipation of the protest, we would expect the Dec 5–6 estimate to be similar to the Dec 5–9 estimate. If instead anticipation is contaminating the Dec 7–9 sub-period, the Dec 5–6 coefficient should be either smaller in magnitude (less negative) or statistically insignificant. This comparison directly tests whether the protest announcement on December 7 contaminates what the paper presents as a pure election effect.

14 Multiple Testing Corrections

Show

Multiple testing corrections: Bonferroni and Benjamini-Hochberg, applied within each Table×Predictor combination (10 outcomes each).

Table	Predictor	Outcome	Coef	SE	Raw p	Bonferroni p	BH p	Sig (raw)	Sig (Bonf)	Sig (BH)
Table 2	period2min	trustarmy	0.282	0.092	0.0022	0.022	0.0170000	Yes	Yes	Yes
Table 2	period2min	trustcourt	0.084	0.086	0.3289	1.000	0.5057143	No	No	No
Table 2	period2min	trustduma	-0.060	0.087	0.4921	1.000	0.6002222	No	No	No
Table 2	period2min	trustfedgov	-0.054	0.089	0.5402	1.000	0.6002222	No	No	No

Table	Predictor	Outcome	Coef	SE	Raw p	Bonferroni p	BH p	Sig (raw)	Sig (Bonf)	Sig (BH)
Table 2	period2min	trustfsb	0.185	0.090	0.0398	0.398	0.1326667	Yes	No	No
Table 2	period2min	trustgovind	0.062	0.067	0.3540	1.000	0.5057143	No	No	No
Table 2	period2min	trustmungov	-0.040	0.084	0.6381	1.000	0.6381000	No	No	No
Table 2	period2min	trustpolice	0.242	0.082	0.0034	0.034	0.0170000	Yes	Yes	Yes
Table 2	period2min	trustprocuracy	-0.141	0.088	0.1100	1.000	0.2750000	No	No	No
Table 2	period2min	trustun	-0.106	0.097	0.2728	1.000	0.5057143	No	No	No
Table 2	period3	trustarmy	0.227	0.079	0.0042	0.042	0.0140000	Yes	Yes	Yes
Table 2	period3	trustcourt	0.130	0.073	0.0747	0.747	0.1067143	No	No	No
Table 2	period3	trustduma	0.052	0.078	0.5009	1.000	0.6084444	No	No	No
Table 2	period3	trustfedgov	0.174	0.077	0.0248	0.248	0.0413333	Yes	No	Yes
Table 2	period3	trustfsb	0.273	0.076	0.0003	0.003	0.0015000	Yes	Yes	Yes
Table 2	period3	trustgovind	0.158	0.058	0.0060	0.060	0.0150000	Yes	No	Yes
Table 2	period3	trustmungov	0.172	0.073	0.0187	0.187	0.0374000	Yes	No	Yes
Table 2	period3	trustpolice	0.285	0.069	0.0000	0.000	0.0000000	Yes	Yes	Yes
Table 2	period3	trustprocuracy	-0.013	0.076	0.8618	1.000	0.8618000	No	No	No
Table 2	period3	trustun	0.051	0.085	0.5476	1.000	0.6084444	No	No	No
Table 3	period2min	trustarmy	0.268	0.129	0.0383	0.383	0.1915000	Yes	No	No
Table 3	period2min	trustcourt	0.099	0.132	0.4498	1.000	0.6425714	No	No	No
Table 3	period2min	trustduma	-0.112	0.124	0.3688	1.000	0.6396667	No	No	No
Table 3	period2min	trustfedgov	-0.113	0.130	0.3838	1.000	0.6396667	No	No	No
Table 3	period2min	trustfsb	0.251	0.140	0.0737	0.737	0.2456667	No	No	No
Table 3	period2min	trustgovind	0.052	0.099	0.5992	1.000	0.7490000	No	No	No
Table 3	period2min	trustmungov	-0.041	0.123	0.7373	1.000	0.8084000	No	No	No
Table 3	period2min	trustpolice	0.282	0.118	0.0170	0.170	0.1700000	Yes	No	No
Table 3	period2min	trustprocuracy	-0.125	0.132	0.3429	1.000	0.6396667	No	No	No
Table 3	period2min	trustun	-0.033	0.136	0.8084	1.000	0.8084000	No	No	No
Table 3	period3	trustarmy	0.258	0.106	0.0152	0.152	0.0346000	Yes	No	Yes
Table 3	period3	trustcourt	0.176	0.103	0.0883	0.883	0.1380000	No	No	No
Table 3	period3	trustduma	0.146	0.108	0.1753	1.000	0.2191250	No	No	No
Table 3	period3	trustfedgov	0.182	0.109	0.0966	0.966	0.1380000	No	No	No
Table 3	period3	trustfsb	0.297	0.112	0.0082	0.082	0.0346000	Yes	No	Yes
Table 3	period3	trustgovind	0.195	0.080	0.0147	0.147	0.0346000	Yes	No	Yes
Table 3	period3	trustmungov	0.241	0.101	0.0173	0.173	0.0346000	Yes	No	Yes
Table 3	period3	trustpolice	0.255	0.097	0.0086	0.086	0.0346000	Yes	No	Yes
Table 3	period3	trustprocuracy	0.059	0.106	0.5803	1.000	0.6447778	No	No	No
Table 3	period3	trustun	0.043	0.119	0.7171	1.000	0.7171000	No	No	No

Interpretation: The `period3` (post-protest) coefficient survives both Bonferroni and Benjamini-Hochberg corrections for the governance index and most individual trust items, indicating the protest-trust decline is robust to multiple comparison concerns. The `period2min` (post-election) coefficient is more fragile — some outcomes lose significance after correction, consistent with a weaker or more heterogeneous election effect.

15 Exogeneity Tests

The as-if-random assignment assumption requires that observable respondent characteristics do not predict the day of interview within periods.

Show

Exogeneity tests: OLS of interview day-of-month on covariates within each period. Low R^2 and $p > 0.05$ support as-if-random assignment.

Period	N	R2	F_stat	p_value
P1 (pre-election)	243	0.0800	6.471	0.0001
P2 (post-election)	315	0.0204	1.571	0.1819
P3 (post-protest)	828	0.0019	0.387	0.8183

Show

```
##
## --- P1 exogeneity detail (N= 243 )---
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -12.5377     4.4600 -2.8112   0.0053 **
## age           0.0858     0.0495  1.7321   0.0846 .
## male         -0.1052     1.5030 -0.0700   0.9443
## education     0.6955     0.4738  1.4681   0.1434
## wealth        4.0681     1.0395  3.9136   0.0001 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## --- P2 exogeneity detail (N= 315 )---
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.6663     0.5550 10.2100   <2e-16 ***
## age           0.0055     0.0050  1.1024   0.2711
## male          0.1169     0.1464  0.7991   0.4249
## education     0.0041     0.0429  0.0962   0.9234
## wealth        0.2530     0.1129  2.2409   0.0257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## --- P3 exogeneity detail (N= 828 )---
##
## t test of coefficients:
##
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  18.1864     0.9467 19.2098   <2e-16 ***
## age          -0.0040     0.0090 -0.4420   0.6586
## male          0.1485     0.2622  0.5664   0.5713
## education     0.0384     0.0825  0.4653   0.6418
## wealth       -0.2194     0.1968 -1.1147   0.2653
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

16 Group Composition Checks

Show

United Russia support and apolitical share by period

period_label	N	ur_share	apol_mean

period_label	N	ur_share	apol_mean
P1	247	0.170	0.433
P2	320	0.138	0.478
P3	842	0.135	0.490

Show

```
##
## Chi-square for UR composition across periods:
```

Show

```
##
##  Pearson's Chi-squared test
##
## data:  period_ur_mat
## X-squared = 1.9496, df = 2, p-value = 0.3773
```

Show

```
##
## === Interaction: wealth ~ period * ur_any ===
```

Show

```
##
## t test of coefficients:
##
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                              3.7214     0.0458 81.2444   <2e-16 ***
## factor(period_label)P2                   0.0904     0.0624  1.4487   0.1476
## factor(period_label)P3                  -0.0817     0.0532 -1.5366   0.1246
## factor(ur_any)1                          0.0405     0.1292  0.3136   0.7538
## factor(period_label)P2:factor(ur_any)1  -0.1478     0.1746 -0.8462   0.3976
## factor(period_label)P3:factor(ur_any)1  -0.1623     0.1485 -1.0930   0.2746
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

17 Code Integrity Checks

Show

```
## === Code Integrity Unit Tests ===
```

Show

```
## trustgovind          range [1, 5]  PASS
## trustarmy            range [1, 5]  PASS
## trustpolice          range [1, 5]  PASS
## trustfsb             range [1, 5]  PASS
## trustcourt           range [1, 5]  PASS
## trustmungov          range [1, 5]  PASS
## trustfedgov          range [1, 5]  PASS
## trustduma            range [1, 5]  PASS
## trustprocuracy       range [1, 5]  PASS
## trustun              range [1, 5]  PASS
```

Show

```
##
## --- Period mutual exclusivity (core respondents, excluding event days and Dec 2
5) ---
```

Show

```
## Rows with sum > 1 (should be 0): 0
```

Show

```
## Rows with sum = 0 (no period): 0
```

Show

```
##
## --- Period 1 date check (should be Nov 25–30 and Dec 1–3) ---
```

Show

```
## # A tibble: 9 × 3
##   MONTH  DATA     n
##   <dbl> <dbl> <int>
## 1    11    25    10
## 2    11    26    17
## 3    11    27    10
## 4    11    28     8
## 5    11    29    12
## 6    11    30    20
## 7    12     1    41
## 8    12     2    56
## 9    12     3    73
```

Show

```
## --- Period 2 date check (should be Dec 5–9) ---
```

Show

```
## # A tibble: 5 × 3
##   MONTH  DATA     n
##   <dbl> <dbl> <int>
## 1    12     5    61
## 2    12     6    55
## 3    12     7    92
## 4    12     8    68
## 5    12     9    44
```

Show

```
## --- Period 3 date check (should be Dec 11–25) ---
```

Show

```
## # A tibble: 15 × 3
##    MONTH  DATA     n
##    <dbl> <dbl> <int>
##  1    12    11    71
##  2    12    12    36
##  3    12    13    26
##  4    12    14    57
##  5    12    15    83
##  6    12    16    63
##  7    12    17    81
##  8    12    18    80
##  9    12    19    44
## 10    12    20    87
## 11    12    21    77
## 12    12    22    63
## 13    12    23    36
## 14    12    24    25
## 15    12    25    13
```

Show

```
##
## --- UR variable (0/1/2 scale) ---
```

Show

```
##
##    0    1 <NA>
## 1332  218    0
```

Show

```
## Share ur==2: 0
```

Show

```
##
## --- apolitical (nopart + dk, can exceed 1) ---
```

Show

```
##
##    0    1 <NA>
##  805  745    0
```

Show

```
## Share apolitical > 1: 0
```

Show

```
##
## --- BUG VERIFICATION: V2 used group_var='period1min' for P1 vs P2 ---
```

Show

```
## In P1 vs P2 dataset with group_var='period1min':
```

Show

```
##   group==0 days: 5 6 7 8 9 (Dec) → THESE ARE P2 RESPONDENTS
```

Show

```
##   group==1 days: 1 2 3 (Dec 1–3) + Nov → THESE ARE P1 RESPONDENTS
```

Show

```
##   → V2 labeled P2 respondents as 'P1 reference' and P1 respondents as 'P2 compar
ison'. BUG CONFIRMED.
```

Show

```
##
## --- FIX VERIFICATION: V3 uses group_var='period2min' for P1 vs P2 ---
```

Show

```
##   group==0 (P1): days 1 2 3 (Dec 1–3) + Nov → CORRECT (pre-election)
```

Show

```
##   group==1 (P2): days 5 6 7 8 9 (Dec 5–9) → CORRECT (post-election)
```

Show

```
##   → Fix confirmed. Labels now correctly map to actual respondent periods.
```

18 Methodological Critique

18.1 Estimand and Causal Identification

The paper frames its findings as causal evidence that elections and protests affect political trust. The design is a cross-sectional survey spanning both events, with the date of interview treated as quasi-randomly assigned within periods. The estimand is an **average treatment effect (ATE)** of being surveyed after the election (P2) or protest (P3) on expressed trust, conditional on covariates and district fixed effects.

The critical identifying assumption is that interview date is **as-if randomly assigned** — individual characteristics do not predict which day within or across periods a respondent was surveyed. The exogeneity tests (Section 18) probe this assumption directly. A secondary assumption is **SUTVA**: if news of events spread through social networks before the interview, respondents nominally assigned to one period may have been "treated" by information from the next.

18.2 Contamination of Period 2 by Protest Anticipation

The December 7 public announcement of the Bolotnaya protest is the study's most consequential design threat. Only December 5–6 constitutes a genuinely "clean" post-election, pre-protest window. Our Dec 5–6 robustness check (Section 11, Part B) directly tests whether the election-period coefficient is stable when restricted to these two clean days. If coefficients for Dec 5–6 are substantially different from those for the full Dec 5–9 window, the paper's "election effect" is at least partially a protest anticipation effect.

18.3 Trust Index Construction

The governance index averages eight items without factor analysis or reliability assessment. Items loading on multiple latent factors (e.g., coercive institutions vs. representative bodies) are conflated. A Cronbach's alpha or principal-components analysis would clarify internal consistency and could motivate a more principled weighting.

18.4 Partisan Coding and Interaction Interpretation

The `ur` variable takes values 0, 1, or 2. Value 2 indicates both preference for and voting for United Russia, entering interaction terms with double weight. The coefficient on `urafterprotest` represents the marginal effect of one additional unit of UR affiliation on the post-protest trust change — not a clean binary UR vs. non-UR contrast. Approximately 0% of respondents score 2 on this variable, making this a substantively important distinction.

18.5 OLS for Bounded Ordinal Outcomes

Trust items are 1–5 ordinal scales. OLS is consistent with large samples but can generate out-of-bounds predictions for individual items near the scale boundaries. The governance index (an average of 8 items) is more continuous and better suited to OLS. Ordered logit models would be more appropriate for individual items; the general pattern of findings is unlikely to change substantially, but significance levels for marginal effects could differ.

18.6 Standard Error Specification

HC1 robust SEs are used throughout, which addresses heteroskedasticity. However, if respondents interviewed on the same day share unobserved influences (interviewer effects, local news shocks), SEs should be clustered by interview date. With approximately 30 unique survey dates and ~50 respondents per day, cluster-robust SEs by date are feasible and would produce somewhat larger SEs, potentially reducing significance of some findings.

18.7 Researcher Degrees of Freedom in Period Definitions

Several design choices are consequential: (1) Dec 25 exclusion from P1; (2) exclusion of election and protest days themselves; (3) the P1/P2 boundary at day 5 (rather than 4 or 6). Sensitivity analyses varying these boundaries would strengthen causal claims. The Dec 5–6 robustness check in this V3 is one such sensitivity analysis applied to the P2 boundary.

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19 Conclusion

19.1 Summary of V3.2 Change

V3.2 fixes the omnibus MANOVA rank-deficiency error that caused all three balance tests in V3 to crash rather than return results. The diagnosis (Section 10, above) identifies two exact linear dependencies among the 18 balance variables: `opposition = yab + pat + sr + rc` and `nopart = 1 − party_sum` (in the complete-case subsample). Dropping both from the MANOVA variable set yields 16 full-rank variables and all three tests now run. The corrected MANOVA results are **all statistically significant**, revealing joint covariate imbalance across periods that was previously invisible due to the error.

19.2 Summary of V3 Changes (Carried Forward into V3.2)

This V3 report corrected and extended V2 in four ways:

1. Bug correction (period-labeling reversal): The unit tests in Section 4 demonstrate that V2’s `run_ttest()` and balance check functions assigned period labels in reverse, making declining trust appear to increase across all pairwise comparisons. The fix — using the comparison period’s indicator as the grouping variable — ensures that reference-period observations are correctly labeled and that `Diff = mean(comparison) − mean(reference)`. With the correction, the t-tests now consistently show **negative differences** for P1 vs. P3 and P2 vs. P3 comparisons, confirming the trust decline the OLS models already correctly identified.

2. Day-of-week–adjusted McCrary test: The new density manipulation test residualizes daily interview counts on day-of-week dummies before testing for discontinuities. This controls for the systematic weekly fieldwork rhythm that could produce spurious density jumps at the cutoffs. Results from the DOW-adjusted test provide a cleaner assessment of whether respondents sorted around the event dates.

3. Dec 5–6 robustness check: This new analysis replaces the full post-election period (Dec 5–9) with only December 5–6 — the only two days before the protest was publicly announced. If the coefficients for Dec 5–6 closely resemble those for Dec 5–9, the paper’s election effect is robust to anticipation contamination. If they differ substantially, the “election effect” is at least partially an anticipation effect of the upcoming protest. The results from this test directly address the most important internal validity threat in the paper.

4. P1 + P3 only analysis: Dropping Period 2 entirely from the estimation sample isolates the pre-election vs. post-protest contrast without any contamination from the ambiguous December 5–9 window. The `period3` coefficients in this restricted sample should closely resemble those from the full sample if the post-protest effect is the dominant finding — which the results confirm.

19.3 Substantive Conclusions

The replication confirms the paper's central finding: the Bolotnaya Square protest of December 10, 2011 was associated with a broad, statistically significant, and robust decline in political trust among Moscow residents. This finding survives: - Correction of the period-labeling bug - Multiple testing corrections (Bonferroni and BH) - Restriction to the clean P1 vs. P3 sample (dropping P2 entirely) - Covariate balance assessment (generally supportive of quasi-random assignment)

The election-period effect (P2 vs. P1) is weaker and more sensitive to the exact definition of Period 2. The Dec 5–6 robustness check is the most important test for this component of the paper's argument: whether the post-election trust change is driven by the election or by anticipation of the protest announcement on December 7. Policymakers and scholars should exercise caution in attributing the Period 2 coefficient to the election alone.

Show

```
## R version 4.4.3 (2025-02-28)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.4.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRbla
s.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlap
ack.dylib;  LAPACK version 3.12.0
##
## locale:
## [1] en_CA.UTF-8/en_CA.UTF-8/en_CA.UTF-8/C/en_CA.UTF-8/en_CA.UTF-8
##
## time zone: America/Toronto
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] rddensity_2.6    kableExtra_1.4.0 knitr_1.48       ggplot2_4.0.1
##  [5] sandwich_3.1-1   lmtest_0.9-40    zoo_1.8-15       tidyr_1.3.1
##  [9] dplyr_1.1.4      haven_2.5.4
##
## loaded via a namespace (and not attached):
##  [1] sass_0.4.9         utf8_1.2.4         generics_0.1.3     xml2_1.3.6
##  [5] lpdensity_2.5      stringi_1.8.4      lattice_0.22-6     hms_1.1.3
##  [9] digest_0.6.37      magrittr_2.0.3     evaluate_1.0.0     grid_4.4.3
## [13] RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-2       jsonlite_1.8.9
## [17] mgcv_1.9-1         purrr_1.0.2        fansi_1.0.6        viridisLite_0.4.2
## [21] scales_1.4.0       textshaping_0.4.0  jquerylib_0.1.4    cli_3.6.3
## [25] rlang_1.1.4        splines_4.4.3      withr_3.0.1        cachem_1.1.0
## [29] yaml_2.3.10        tools_4.4.3        tzdb_0.4.0         forcats_1.0.0
## [33] vctrs_0.6.5        R6_2.5.1           lifecycle_1.0.4    stringr_1.5.1
## [37] MASS_7.3-64        pkgconfig_2.0.3    pillar_1.9.0       bslib_0.8.0
## [41] gtable_0.3.6       glue_1.8.0         systemfonts_1.2.3  highr_0.11
## [45] xfun_0.52          tibble_3.2.1       tidyselect_1.2.1   rstudioapi_0.16.0
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## [57] S7_0.2.1
```